

Design by Optimization of an Axial-Flux Permanent-Magnet Synchronous Motor Using Genetic Algorithms

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Abstract — In this paper, a genetic algorithm (GA) is used to optimize a motor for an automotive application. This motor is a double stator axial permanent-magnet (PM) machine, and the optimization uses a simplified analytical model. The solution obtained with the tool is compared to three-dimensional (3-D) finite-elements method (FEM) to confirm the validity of the model.

Keywords — Optimization, genetic algorithms, axial-flux PM synchronous motor, embedded PM, analytical model.

NOMENCLATURE

μ_0 , vacuum permeability.	D_{int} , internal diameter.
μ_r , relative permeability.	D_{ext} , external diameter.
B_m , PM flux density.	R_m , average radius of the air-gap.
H_m , PM magnetization.	e_g , air-gap depth.
B_r , PM remanent flux density.	S_g , two poles air-gap surface.
B_g , air-gap flux density.	L_{pm} , PM length in the axial direction.
B_{st} , stator yoke flux density.	e_{pm} , PM thickness.
B_t , stator tooth flux density.	d_{sl} , slot depth.
p , number of pole pairs.	L_{st} , stator yoke thickness.
N_{sl} , number of slots.	M_{pm} , PM weight.
n , number of slots per phase.	L_{mot} , axial motor length.
K_s , linear current density.	T_{nom} , nominal torque.
J_s , current density	

I. INTRODUCTION

Considering automotive applications, the electric machines have severe constraints in terms of size and weight. Due to its high torque density [1-2], a double stator axial-flux PM synchronous motor (AFPMSM) is chosen. To satisfy the complex specifications (in particular in terms of specific torque and specific power), a design by optimization using an analytical model and a genetic algorithm (NSGA-II) is performed [3]. The GAs enable to solve multi-objective

optimization problems. The advantage of multi-objective optimization is to obtain a Pareto Front of optimal solutions for several optimization criteria (such as total weight, PM weight, maximal torque, efficiency...). The choice of the solution can thus be a compromise between different optimization criteria. Design by optimization of axial-flux PM machines using GAs is not widely studied. Axial-flux surface-mounted PM machines, for their simple analytical design, comparatively to axial-flux embedded PM, are mostly studied. In [4], the authors studied a design of a 10 kW AFPMS Generator (AFPMSG) with two rotors surface-mounted PM and internal stator using GA optimization. Electromagnetic, thermal and mechanical models are used in the optimization process. The multi-objective optimization function consists in minimizing the PM volume and maximizing the efficiency. In [5], the machine studied is a 30 kW AFPMSG with two stators and rotor surface-mounted PMs. In this case, the objective function is only the cost of the active materials. The authors in [6] applied different optimization methods to design AFPMSM with two rotors surface-mounted PM and internal stator. The results show that stochastic algorithms are more interesting than deterministic algorithms when applied to complex problems. The design of electric machines can be a combination between optimization design using a simple analytical model based on sizing equation and FEM. In [7], an AFPMSM with internal stator and two external surface-mounted PM rotors is studied. The authors used GA optimization to obtain motor dimensions and 3-D FEM to obtain motor characteristics.

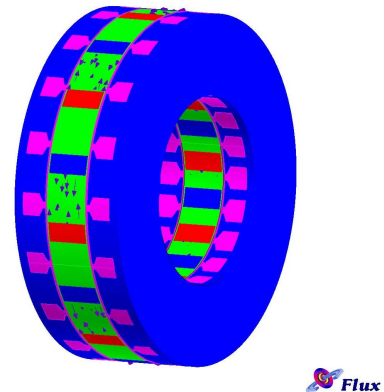


Fig. 1. Axial-flux embedded PM synchronous machine.

Using FEM, motor parameters are re-optimized to improve the machine performances, such as reduction of cogging torque by PM skewing and modification of winding configuration.

In this paper, the novelty is that the machine considered, for the multi-objective optimization, is an axial-flux synchronous machine with embedded PM, as shown in Fig. 1. To achieve the design, a simplified analytical model of the motor is used. The steps of the algorithm NSGA-II, the variation range of the design parameters and the optimization constraints are described. Then, the optimization results are discussed for the considered multi-objective function. Finally, to verify the validity of the analytical model, the solution obtained with the optimization is simulated using 3-D FEM.

II. ANALYTICAL MODEL

The target of this simplified analytical model is to express the torque per unit of weight. Then, firstly, the flux density in the PM can be expressed by:

$$B_m = B_r + \mu_0 \cdot \mu_r \cdot H_m \quad (1)$$

By applying the Ampere's law to the elementary magnetic circuit represented in Fig. 2, we can write:

$$H_m \cdot e_{pm} + 2 \cdot H_g \cdot e_g = -\frac{2\pi \cdot R_m \cdot K_s}{N_{sl}} \quad (2)$$

By considering the flux conservation in the magnetic circuit, we can write:

$$B_m \cdot \frac{L_{pm}}{2} \cdot \frac{D_{ext} - D_{int}}{2} = B_g \cdot \frac{S_g}{2} \quad (3)$$

Using (1), the PM magnetization is given by:

$$H_m = \frac{B_m - B_r}{\mu_0 \cdot \mu_r} \quad (4)$$

From (1)-(4), the expressions of the magnetic flux density in the PM, B_m , and the air-gap, B_g , are given by:

$$B_m = \frac{\frac{B_r}{\mu_0 \mu_r} \cdot e_{pm} - \frac{2\pi \cdot R_m \cdot K_s}{N_{sl}}}{\frac{1}{\mu_0 \mu_r} \cdot e_{pm} + 2 \cdot e_g \cdot \frac{L_{pm} \cdot \frac{D_{ext} - D_{int}}{2}}{S_g}} \quad (5)$$

$$B_g = \frac{\frac{B_r}{\mu_0 \mu_r} \cdot e_{pm} - \frac{2\pi \cdot R_m \cdot K_s}{N_{sl}}}{\frac{1}{\mu_0 \mu_r} \cdot e_{pm} + 2 \cdot e_g \cdot \frac{L_{pm} \cdot \frac{D_{ext} - D_{int}}{2}}{S_g}} \cdot \frac{L_{pm} \cdot \frac{D_{ext} - D_{int}}{2}}{S_g} \quad (6)$$

The expression of S_g , the section of the air-gap seen by the rotor yoke between two poles, as shown in Fig. 3, is given by:

$$S_g = \left(\frac{D_{ext} - D_{int}}{2} \right) \left[\frac{\pi \cdot (D_{ext} + D_{int})}{4p} - e_{pm} \right] \quad (7)$$

And finally, the electromagnetic torque of the motor can be defined as:

$$T_{nom} = \alpha \cdot B_g \cdot K_s \cdot \left(\frac{D_{ext} + D_{int}}{2} \right)^2 \cdot (D_{ext} - D_{int}) \quad (8)$$

where α is a constant, which depends on the winding factor.

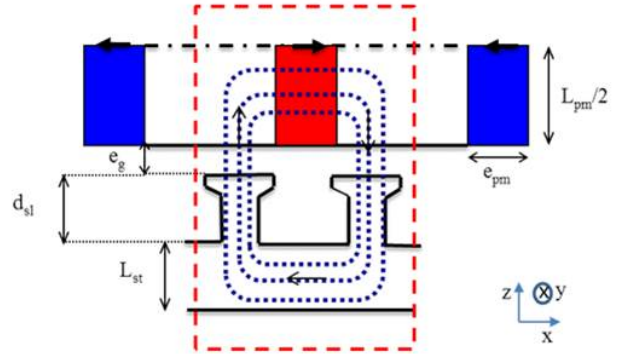


Fig. 2. Circulation of the flux produced by one PM pole.

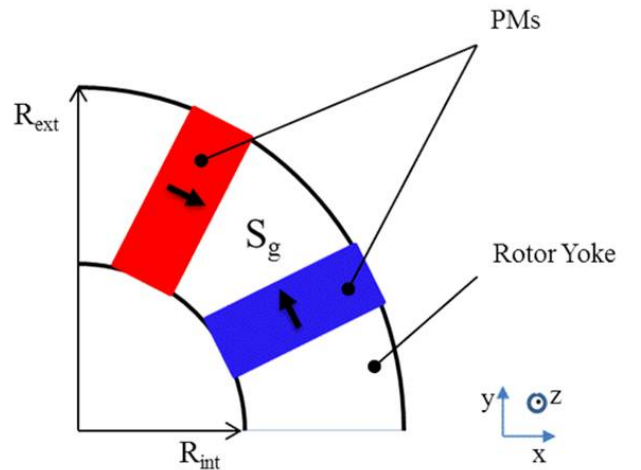


Fig. 3. Representation of one pair of poles according to the plan (xoy).

III. FORMULATION OF THE OPTIMIZATION PROBLEM

The genetic algorithm used for the optimization is the NSGA-II algorithm described in [3]. The different optimization steps of the algorithm are shown in Fig.4.

First, an initial population is randomly generated. Then, the fitness of each individual (candidate solution) is evaluated. If the required fitness is achieved, the individuals are saved. Otherwise, the best individuals, according to their fitness, are selected and modified to form a new population, which is used in the next iteration. The modification of the individuals consists in mutation and recombination operations of the chromosomes. The new population is then evaluated and the same operations are repeated until the target fitness level is achieved or the maximum number of generations is reached.

The chromosomes of the individuals consist on design parameters. In this study, external diameter, internal diameter, stator yoke thickness, PM thickness, PM axial length, numbers of slots per phase and slot depth are selected as chromosomes. The constraints concern air-gap, stator yoke and tooth flux densities, PM weight, motor axial length, nominal torque and PM magnetization. Then, the optimization problem has seven optimization parameters; six are continuous and one is discrete, as shown in the Table 1.

More precisely, the geometric and electromagnetic constraints are defined in Table 2. The multi-objective function consists in maximizing the torque density and minimizing the PM weight.

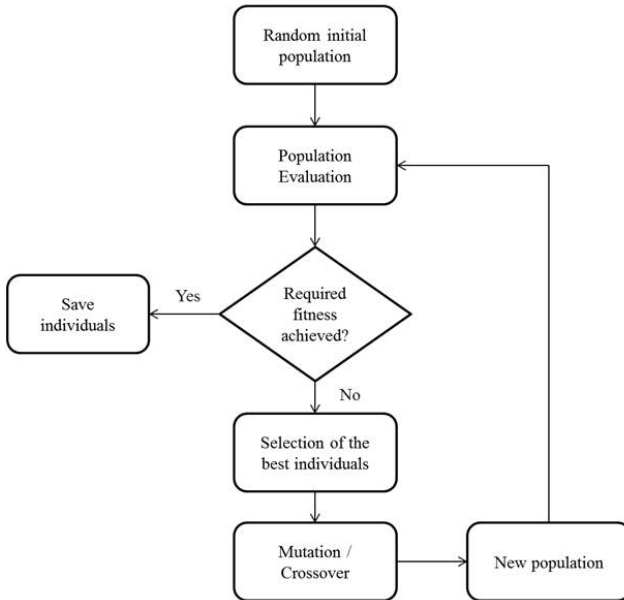


Fig. 4. Representation of the GA optimization steps.

TABLE 1
DOMAIN OF VARIATION OF THE OPTIMIZATION PARAMETERS

Design Parameters	Symbols	Units	Type	Variation Range
External Diameter	D_{ext}	[mm]	Continuous	[180, 230]
Internal Diameter	D_{int}	[mm]	Continuous	[120, 160]
Stator Yoke Thickness	L_{st}	[mm]	Continuous	[10, 30]
PM Thickness	e_{pm}	[mm]	Continuous	[1, 50]
PM Axial Length	L_{pm}	[mm]	Continuous	[1, 50]
Number of Slots per Phase	n	-	Discrete	2,3...8
Slot Depth	d_{sl}	[mm]	Continuous	[5,50]

TABLE 2
THE OPTIMIZATION CONSTRAINTS

Optimization Constraints	Symbols	Units	Min Value	Max Value
Airgap flux density	B_g	[T]	0.6	0.9
Stator yoke flux density	B_{st}	[T]	0.7	1.7
Stator tooth flux density	B_t	[T]		1.8
PM weight	M_{pm}	[kg]	0.8	1.3
Motor axial length	L_{mot}	[mm]	-	102
Nominal torque	T_{nom}	[N.m]	55	-
PM magnetization	H_m	[A/m]	H_{cj}	-

TABLE 3
OPTIMIZATION TOOL PARAMETERS

Number of individuals	100
Number of generations	100
Number of chromosomes	7
Number of constraints	7
Number of criteria	2

IV. RESULTS OF THE OPTIMIZATION

The number of individuals and the number of generations are defined in Table 3. To save a solution, it is necessary to satisfy all constraints.

In Figure 5, the solutions obtained with the optimization show that the maximization of the torque density can be

achieved by minimizing the PM weight if the best solutions are chosen. The constraint on the axial length of the machine and on the rated torque involves very different dimensions of the PMs. Thus, as shown in Fig. 6-7, if the PM axial length decreases, the PM thickness must increase to satisfy the torque constraint. The external diameter increases also, which increases the PM weight and the total machine weight. Thus, the torque density decreases when PM weight becomes higher. Otherwise, if the axial length of the machine increases, the ratio between the air-gap surface seen by one pole and the surface of one pole becomes larger. Thus, the torque performance of the machine is improved. In addition, the motor weight decreases, which increases the torque density. Thereby, the proposed tool can help to find the best solution for a minimum PM weight and a maximum torque density.

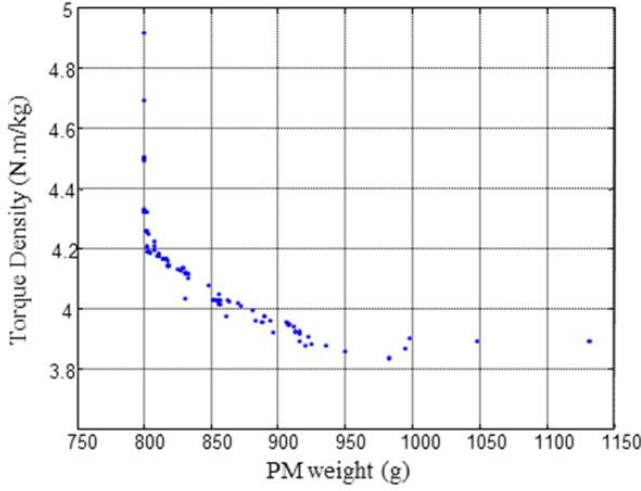


Fig. 5. Evolution of the optimization criteria.

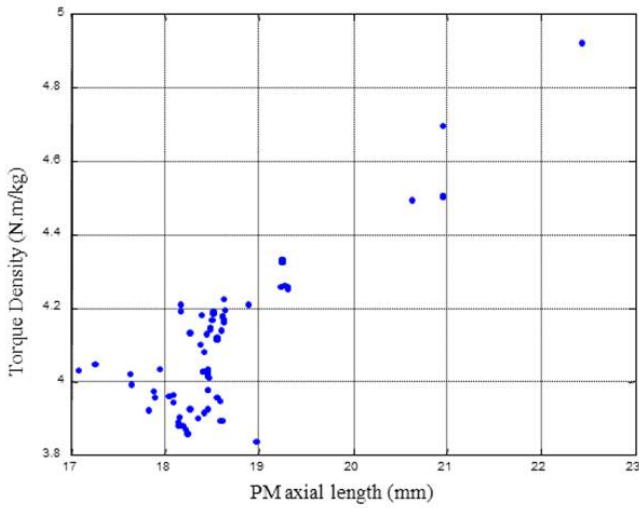


Fig. 6. Torque density as function of PM axial length.

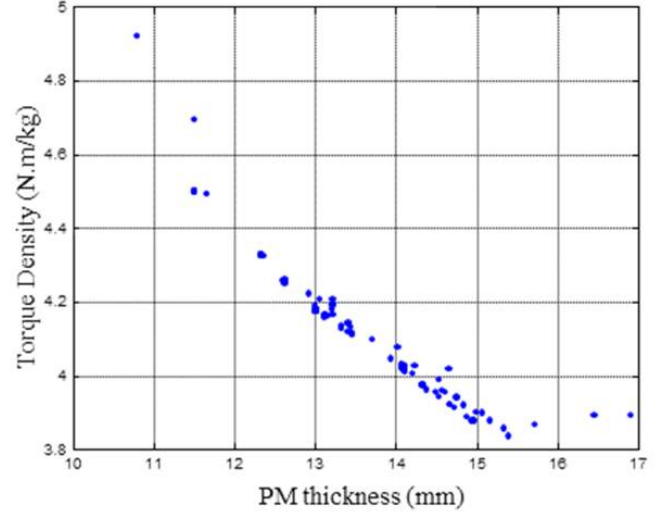


Fig. 7. Torque density as function of PM thickness.

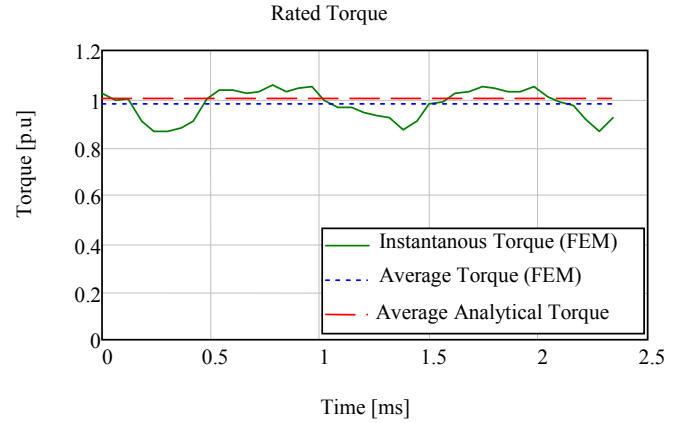


Fig. 8. Torque as function of the time.

V. 3-D FINITE-ELEMENT METHOD

The optimal solution obtained with the design by optimization is simulated using a 3-D FEM software [8]. The evolution of the motor torque is illustrated in Fig. 8 (for the rated operating point).

The error in the calculation of the nominal torque by the analytical model, compared to the average torque obtained with 3-D FEM, is less than 5 %. It confirms the reliability of the developed approach.

VI. CONCLUSION

In this work, the authors have proposed the optimal design of an original motor topology, consisting in an axial-flux synchronous machine with embedded PM. The optimal design is based on a simplified analytical model combined with a genetic algorithm. The analytical model is as simple as possible in order to improve the convergence speed, unless we use a stochastic optimization algorithm. The advantage of the optimization based on GAs is to reach a global optimum and to follow the evolution of solutions based on design parameters. However, the GAs are all the more interesting when the model is complex and multi-physic (electromagnetic,

thermal and mechanical models) and when the parameters are numerous and hybrid (i.e. continuous, discrete and possibly Boolean). However, this first order approach is satisfying in the sense that it enables to provide an interesting design that is confirmed by 3D FEM refined model.

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